

OMNISHARES

Complete Product & Technical Specification

Your Complete Investment Intelligence Platform

Build a complete, production-quality investment platform called OMNISHARES.

OMNISHARES is an AI-powered investment intelligence and portfolio-management platform, starting with the Kenyan Nairobi Securities Exchange (NSE), with architecture designed to expand to other African and global markets later.

The platform must feel like a serious financial product—not a simple stock-price website.

DESIGN PHILOSOPHY

Combine:

- Bloomberg-style financial information
- TradingView-style charts
- Modern fintech UX
- Apple-level simplicity
- Clean professional dark interface
- Mobile-first responsive design

Primary currency:

KES (KSh)

Primary market:

Nairobi Securities Exchange (NSE)

The system must be designed so mock/demo data can be replaced with verified real market data without rebuilding the frontend.

1. BRAND

Name:

OmniShares

Tagline:

"Your Complete Investment Intelligence Platform"

Alternative short description:

"Research. Track. Understand. Invest."

Create a premium financial identity.

Use:

- Dark professional interface
- Clean typography
- High information density without clutter
- Rounded cards
- Subtle borders
- Elegant charts
- Clear green/red financial indicators
- KSh formatting
- Professional icons
- Responsive mobile navigation

Do NOT make it look like a gambling/trading app.

The product should encourage long-term investing and informed decisions.

2. APPLICATION STRUCTURE

Main navigation:

1. Dashboard

2. Markets

3. Companies

4. Portfolio

5. Sectors

6. Screener

7. Watchlist

8. Alerts

9. News

10. OmniAI

11. OmniScore

12. Settings

On mobile, use a bottom navigation bar for the most important sections and a menu for secondary sections.

3. DASHBOARD

The dashboard is the command centre.

At the top display:

Portfolio Value

Invested Capital

Cash Balance

Today's P/L

Total P/L

Total Return %

Dividend Income

Example:

Portfolio Value

KSh 25,430

Today's Change

+KSh 320 (+1.27%)

Total Profit

+KSh 2,430 (+10.56%)

Dividend Income

KSh 840

Create an interactive portfolio performance chart.

Chart periods:

1D

1W

1M

3M

6M

1Y

5Y

MAX

Below the portfolio chart display:

MARKET OVERVIEW

- NASI
- NSE 20
- NSE 25
- Market sentiment
- Market volume
- Market status

Then:

TOP GAINERS

TOP LOSERS

MOST ACTIVE

TOP DIVIDEND STOCKS

Then:

MY PORTFOLIO

Display:

Company

Ticker

Shares

Current price

Market value

Today's change

Total P/L

Then:

PORTFOLIO ALLOCATION

Visualize allocation by:

- Company
- Sector
- Asset type

Then:

RECENT ACTIVITY

Show:

- Purchases
- Sales
- Dividends
- Deposits
- Withdrawals

Then:

ALERTS

Show the latest important alerts.

Then:

OMNISHARES INSIGHT

Provide a short AI-generated portfolio insight.

Example:

"Your portfolio is currently concentrated in banking, which represents 46% of your holdings. Consider increasing exposure to other sectors as your portfolio grows."

4. MARKETS

Create a dedicated Markets page.

Sections:

Market Summary

Market Indices

Gainers

Losers

Most Active

Highest Dividend Yield

52-Week Highs

52-Week Lows

Allow users to search NSE-listed companies.

Each stock row should show:

Company

Ticker

Price

Change

Change %

Volume
Market Cap
P/E
Dividend Yield
OmniScore

5. COMPANY DIRECTORY

Create a complete company directory.

Allow filtering by:

Sector
Market Cap
Price
Dividend Yield
P/E
OmniScore
Performance

Company cards/table should contain:

Company name
Ticker
Current price
Daily change
Sector
Market cap
Dividend yield
OmniScore

6. COMPANY PROFILE PAGE

Every company must have a detailed profile.

Example:

CO-OPERATIVE BANK OF KENYA

Ticker: COOP
Display:
Current Price
Daily Change
Market Cap
P/E
EPS
Dividend Yield
52-Week High
52-Week Low

PRICE CHART

Periods:

1D
1W
1M
6M
1Y
5Y
MAX

FUNDAMENTALS

Revenue
Net Profit
EPS
ROE
ROA
Debt
Assets
Cash
Free Cash Flow

DIVIDENDS

Show:
Dividend history
Dividend per share
Dividend yield
Payment dates
Ex-dividend dates

VALUATION

Show:
P/E
P/B
PEG where available
Historical valuation
Estimated fair-value range where supported

RISK

Show:
Volatility
Debt risk
Earnings risk
Sector concentration
Market risk

OMNISCORE

Display a score from 0–100.

7. OMNISCORE

Create OmniScore™, a proprietary stock-analysis scoring framework.

Score factors:

Value — 20%

Growth — 20%

Financial Health — 20%

Dividend — 15%

Risk — 10%

Momentum — 10%

Management — 5%

Total = 100%

Display:

OMNISCORE

82/100

Breakdown:

Value 88

Growth 76

Health 91

Dividend 84

Risk 70

Momentum 79

Management 82

Do not make OmniScore an automatic "buy" signal.

Instead show:

Strong

Positive

Neutral

Caution

Weak

Explain why the score was generated.

8. MY PORTFOLIO

This is one of the most important sections.

Create a complete personal investment tracker.

Portfolio overview:

Total Value

Invested Capital

Cash

Total Profit/Loss

Return %

Today's P/L

Dividend Income

HOLDINGS

Columns:

Company

Ticker

Shares

Average Buy Price

Current Price

Invested Amount

Market Value

Profit/Loss

Return %

Dividend Income

Sector

Allow sorting.

9. TRANSACTIONS

Create complete transaction history.

Transaction types:

BUY

SELL

DIVIDEND

DEPOSIT

WITHDRAWAL

FEE

OTHER

Each transaction contains:

Date

Transaction type

Company

Shares

Price

Fees

Total amount

Notes

Allow:

Search

Filter

Sort

Export

10. BUY TRANSACTION

Create a simple form:

Company

Number of shares

Buy price

Transaction date

Broker

Fees

Notes

Automatically calculate:

Gross amount

Total cost

Average cost

Portfolio impact

11. SELL TRANSACTION

Fields:

Company

Shares sold

Sell price

Date

Broker

Fees

Automatically calculate:

Sale value

Realized profit/loss

Return %

Remaining shares

12. DIVIDENDS

Create a dedicated Dividend section.

Display:

Total dividends

This year

Last year

Monthly dividend income

Dividend yield on cost

Dividend history:

Company

Shares held

Dividend per share

Gross dividend

Tax

Net dividend

Payment date

Create a dividend-income chart.

13. DEPOSITS AND WITHDRAWALS

Track money entering and leaving the portfolio.

DEPOSIT

Date

Amount

Method

Reference

Notes

WITHDRAWAL

Date

Amount

Method

Reference

Notes

Show:

Total deposited

Total withdrawn

Net contributions

14. PROFIT / LOSS

Create a dedicated performance page.

Show:

Unrealized P/L

Realized P/L

Dividend income

Fees

Net return

Charts:

Portfolio value over time

Profit/loss over time

Contributions vs portfolio value

Dividend income over time

Allow:

1M

3M

6M

1Y

5Y

MAX

15. PORTFOLIO ALLOCATION

Create visual allocation charts.

BY COMPANY

COOP — 30%

KCB — 25%

KEGN — 20%

Safaricom — 15%

Other — 10%

BY SECTOR

Banking

Energy

Telecommunications

Insurance

Manufacturing

Investment

Other

Display concentration warnings.

Example:

"Banking represents 48% of your portfolio."

"Your portfolio is highly concentrated in one sector."

16. WATCHLIST

Allow users to add stocks to a watchlist.

Each item shows:

Company

Price

Daily change

OmniScore

Dividend yield

P/E

Allow custom price alerts.

17. ALERTS

Create a powerful notification system.

Users can create:

PRICE ALERT

Example:

Notify me when KCB reaches KSh 50.

PERCENTAGE ALERT

Notify me if a stock rises/falls more than 5%.

DIVIDEND ALERT

Notify me when:

- Dividend is announced
- Ex-dividend date approaches
- Dividend is paid

EARNINGS ALERT

Notify when company results are released.

OMNISCORE ALERT

Notify when OmniScore changes significantly.

PORTFOLIO ALERTS

Notify when:

- Portfolio gains 10%
- Portfolio loses 5%
- Portfolio reaches target value
- Sector concentration exceeds chosen limit

TRANSACTION ALERTS

Notify after:

- Buy
- Sell
- Deposit
- Withdrawal
- Dividend

NEWS ALERTS

Notify about major developments affecting watched companies.

Create an Alerts Centre with:

Unread

Read

Important

Price

Dividend

Portfolio

News

18. NEWS

Create an NSE-focused financial news page.

Categories:

NSE

Companies

Banking

Energy

Insurance

Telecom

Economy

CBK

Government policy

Corporate results

Each article:

Headline

Source

Date

Company

Summary

Use AI to summarize long articles.

19. STOCK SCREENER

Create a powerful stock screener.

Filters:

Price

Market Cap

P/E

P/B

EPS

Revenue growth

Profit growth

ROE

ROA

Debt

Dividend yield

Dividend growth

OmniScore

Sector

Price performance

Example query:

Dividend yield > 5%

P/E < 12

ROE > 12%

OmniScore > 70

Display matching companies.

20. SECTORS

Create sector dashboards.

For each sector show:

Sector performance

Companies

Market cap

Average P/E

Average dividend yield

Top performers

Worst performers

Allow users to compare sectors.

Portfolio section should use the same sector classification.

21. OMNI AI

Create an AI investment assistant called OmniAI.

Users can ask:

"Should I buy COOP?"

"Compare KCB and Equity."

"Which stocks have strong dividends?"

"Is my portfolio diversified?"

"What sector am I overexposed to?"

"Explain why my portfolio is down."

"Find undervalued NSE companies."

OmniAI should analyze available verified data.

Responses should include:

Summary

Strengths

Weaknesses

Risks

Valuation

Dividend profile

Portfolio impact

Conclusion

Never present AI output as guaranteed financial advice.

22. PORTFOLIO AI

OmniAI should analyze the user's personal portfolio.

Examples:

"How diversified am I?"

"What is my best-performing stock?"

"Which sector is dominating?"

"How much dividend income am I generating?"

"How much have I invested?"

"What is my average annual return?"

"Which holdings are causing most of my losses?"

Provide clear explanations and charts where appropriate.

23. INVESTMENT GOALS

Allow users to create goals.

Example:

Goal:

Build KSh 1,000,000 portfolio.

Display:

Current:

KSh 245,000

Target:

KSh 1,000,000

Progress:

24.5%

Estimated timeline based on contribution assumptions.

Other goals:

Dividend income goal

Retirement goal

Emergency investment goal

Long-term wealth goal

24. USER ACCOUNT

Create authentication.

Users can:

Register

Login

Logout

Reset password

Profile:

Name

Email

Phone

Currency

Investment preferences

Security:

Password hashing

Secure authentication

Session management

Input validation

25. SETTINGS

Settings:

Profile

Security

Notifications

Alert preferences

Currency

Theme

Data preferences

Privacy

Export data

Allow:

Light mode

Dark mode

System mode

Default:

Dark mode.

26. DATA MODEL

Core database entities:

users

portfolios

holdings

transactions

deposits

withdrawals

dividends

companies

prices

financials

sectors

news

watchlists

alerts

omniscores

portfolio_snapshots

investment_goals

notifications

settings

Relationships must be properly normalized.

27. MARKET DATA

Build a market-data abstraction layer.

Do NOT hard-code the frontend to a single provider.

Structure:

Market Data Provider

↓

Data Normalization Layer

↓

Database

↓

API

↓

Frontend

This allows future replacement of data providers.

Where real-time data is unavailable, clearly label data as:

Delayed

End-of-day

Demo

Unavailable

Never fabricate real market prices.

28. API

Create REST API endpoints for:

Authentication

Users

Companies

Prices

Financials

Portfolio

Holdings

Transactions

Dividends

Deposits

Withdrawals

Alerts

Watchlists

News

OmniScore

AI

Sectors

Screener

Use proper:

Authentication

Authorization

Validation

Error handling

Rate limiting

Logging

29. RESPONSIVE DESIGN

The website must work beautifully on:

Android

iPhone

Tablet

Laptop

Desktop

Mobile experience is extremely important.

Desktop:

Sidebar navigation.

Mobile:

Bottom navigation plus menu.

Charts must resize automatically.

Tables must become scrollable cards on small screens.

30. SECURITY

Implement:

HTTPS-ready architecture

Password hashing

JWT/session security

Input validation

SQL injection protection

XSS protection

CSRF protection where applicable

Role-based access

Secure API keys

Environment variables

Audit logging

Never expose API keys in frontend code.

31. ADMIN DASHBOARD

Create an administrator dashboard.

Admin can manage:

Users

Companies

Market data

News

Alerts

OmniScore configuration

Sectors

System health

API connections

Show:

Active users

Registered users

System status

Data freshness

API health

Errors

32. EXPORT

Allow users to export:

Portfolio

Transactions

Dividends

Profit/loss

Deposits

Withdrawals

Formats:

CSV

Excel

PDF

33. FUTURE BROKER INTEGRATION

Design the architecture so future broker integrations can be added.

Potential future workflow:

OmniShares

↓

Broker connection

↓

Account

↓

Order

↓

Confirmation

↓

Portfolio update

DO NOT implement automatic trading unless a secure, compliant broker API is available.

All trade execution must require explicit user confirmation.

34. FUTURE ETF SUPPORT

Although the first market is NSE, design the asset architecture to eventually support:

Kenyan stocks

African stocks

Global stocks

ETFs

Bonds

REITs

Other regulated assets

Do not hard-code the system to equities only.

35. PERFORMANCE

Optimize for:

Fast loading

Lazy loading

Caching

Pagination

Efficient database queries

Chart optimization

API caching

Dashboard should feel fast even with thousands of transactions.

36. ERROR STATES

Never show broken UI.

Handle:

No data

Market closed

API unavailable

Data delayed

Network error

Invalid transaction

Insufficient information

Use friendly messages.

Example:

"Market data is temporarily unavailable. Your portfolio records remain safe."

37. DEMO MODE

The application should initially work using realistic demo data.

Demo data must be clearly labelled.

The user should be able to explore:

Dashboard

Markets

Company pages

Portfolio

Charts

Alerts

Watchlist

AI

Screener

without needing real financial connections.

38. IMPORTANT FINANCIAL RULE

Never invent market data.

If real data is unavailable:

Say:

"Data unavailable"

or:

"Demo data"

Do not present fictional prices, dividends or financial results as real.

39. VISUAL DASHBOARD

The dashboard should visually resemble a premium financial command centre.

Layout:

OMNISHARES Search Profile

Portfolio Value

KSh 25,430

+1.27% Today

[Portfolio Performance Chart]

1D 1W 1M 3M 6M 1Y 5Y MAX

Portfolio Profit/Loss Dividends

KSh 25,430 +KSh 2,430 KSh 840

MARKET OVERVIEW

NASI NSE20 NSE25

TOP GAINERS TOP LOSERS

Safaricom Company X

COOP Company Y

KCB Company Z

MY HOLDINGS

Company Shares Value P/L Return

COOP 20

KCB 10

KEGN 15

PORTFOLIO ALLOCATION

[Pie / Donut Chart]

Banking 40%

Energy 25%

Telecom 15%

Insurance 10%

Other 10%

RECENT ACTIVITY

BUY COOP

DIVIDEND KCB

BUY KEGN

ALERTS

- ■ Banking concentration above 40%
- COOP dividend announced
- KCB reached target price

OMNISHARES AI

"Your portfolio is performing positively, but banking accounts for a large percentage of your holdings."

[Ask OmniAI]

40. USER EXPERIENCE PRINCIPLE

The user should be able to open OmniShares and understand their financial position within 10 seconds.

The most important information must always be visible.

Avoid unnecessary complexity.

Use progressive disclosure:

simple first,

advanced information when requested.